PSEUDO-COHERENCE DESCENDS ALONG AN FPPF COVERING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Pullback preserves pseudo-coherence, which gives the easy direction. invoked at proof lines 3078-3080. Settles one implication of the statement.
- Finiteness of a module over an affine base descends along a surjective flat morphism of finite presentation. invoked at proof lines 3095-3105. Produces a finite type module on the base whose pullback surjects onto the top cohomology.
- In a distinguished triangle, if two terms are pseudo-coherent then so is the third, which lets the argument reduce the degree. invoked at proof lines 3105-3115. Keeps the cone pseudo-coherent so that the induction on the degree can proceed.

Source reference. Original lemma 09U7: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 3078-3080 for the easy implication; the converse, which is the substance, is inside the counted span.

Source reference. Original lemma 09UB: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 3095-3102 to descend finiteness of the top cohomology; the induction argument that consumes it is inside the counted span.

Source reference. Original lemma 08DN: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 3096-3100 to see that the top cohomology of the pulled back complex is of finite type; the descent step built on it is inside the counted span.

Source reference. Original lemma 08CD: sources/stacks-chow/tags.txt, source lines 1–1. Applied at proof lines 3108-3115 to transfer the property along the triangle and conclude m-pseudo-coherence of the original complex; the construction of the triangle is inside the counted span.

Lemma 0.2. *Let $\{f_i : X_i \to X\}$ be an fppf covering of schemes. Let $E \in D(\mathcal{O}_X)$. Let $m \in \mathbf{Z}$. Then E is m-pseudo-coherent if and only if each Lf_i^*E is m-pseudo-coherent.*

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Proof. Pullback always preserves m-pseudo-coherence, see Cohomology, Lemma 09U7. Conversely, assume that Lf_i^*E is m-pseudo-coherent for all i. Let $U \subset X$ be an affine open. It suffices to prove that $E|_U$ is m-pseudo-coherent. Since $\{f_i : X_i \to X\}$ is an fppf covering, we can find finitely many affine open $V_j \subset X_{a(j)}$ such that $f_{a(j)}(V_j) \subset U$ and $U = \bigcup f_{a(j)}(V_j)$. Set $V = \coprod V_i$. Thus we may replace X by U and $\{f_i : X_i \to X\}$ by $\{V \to U\}$ and assume that X is affine and our covering is given by a single surjective flat morphism $\{f : Y \to X\}$ of finite presentation.

Since f is flat the derived functor Lf^* is just given by f^* and f^* is exact. Hence $H^i(Lf^*E) = f^*H^i(E)$. Since Lf^*E is m-pseudo-coherent, we see that $Lf^*E \in D^-(\mathcal{O}_Y)$. Since f is surjective and flat, we see that $E \in D^-(\mathcal{O}_X)$. Let $i \in \mathbf{Z}$ be the largest integer such that $H^i(E)$ is nonzero. If $i < m$, then we are done. Otherwise, $f^*H^i(E)$ is a finite type $\mathcal{O}_Y$-module by Cohomology, Lemma 08DN. Then by Descent, Lemma 09UB the $\mathcal{O}_X$-module $H^i(E)$ is of finite type. Thus, after replacing X by the members of a finite affine open covering, we may assume there exists a map

$$\alpha : \mathcal{O}_X^{\oplus n}[-i] \longrightarrow E$$

such that $H^i(\alpha)$ is a surjection. Let C be the cone of α in $D(\mathcal{O}_X)$. Pulling back to Y and using Cohomology, Lemma 08CD we find that Lf^*C is m-pseudo-coherent. Moreover $H^j(C) = 0$ for $j \geq i$. Thus by induction on i we see that C is m-pseudo-coherent. Using Cohomology, Lemma 08CD again we conclude. □